

Training Step

A training step is one update to the model's weights

In one training step

1. **Take a batch of data:** a small chunk of the training set (e.g., 32 or 256 examples), not the whole dataset at once
 2. **Forward pass:** run that batch through the model to get prediction
 3. **Compute the loss:** measure how strong the prediction was
 4. **Backward pass:** compute the gradients of the loss with respect to every weight (backpropagation)
 5. **Update the weights:** the optimizer (SGD, Adam, ...) nudges the weights slightly in the direction that reduces the loss, scaled by the learning rate.
- **Batch size:** how many examples are used in one step
 - **Step (iteration):** one weight update, using one batch
 - **Epoch:** one full pass through the entire training dataset.

Example:

- 10,000 training samples
 - Batch size = 500
 - 10 epochs
 - Warmup steps = 100
- Then:**
- Step per epoch = $10,000 / 500 = 200$ steps
It takes 200 steps to get through all 10,000 examples once
 - total step = $10 \times 200 = 2000$ steps
 - warmup step = 5% of the whole training steps 2000